MXB103 Derivations

Every formula in the unit, derived slowly, one step at a time

Every derivation in this document has the same layout:

- **Goal**: the formula we are about to prove.
- **Idea**: the one-sentence plan, so you know where the steps are heading.
- **Steps**: each one says *what* to do and *why*. When the step matches an official marking scheme, its marks are shown in [red].
- **Result**: the final formula in a green box.

Priority stars

★★★	Asked in the 2023, 2024 or 2025 exam. **Learn to write it from memory.**
★★	A core building block the exam relies on, or a likely variation.
★	For understanding. Not asked directly, but it explains *why* a method works.

How to practise. Cover everything below the Goal, write the derivation on blank paper, then compare step by step. A derivation is "learnt" when you can do it twice in a row without looking.

★★★ The exam derivations (learn these first)

D2.1	Cancellation-free quadratic formula	2025 Q1b (3 marks)
D2.4	Taylor polynomial + error bound (full layout)	Q2 every year (6 marks)
D4.1	Total derivative $f'(t, y)$	needed for every Taylor 2 question
D4.2	Second-order Taylor method	2023 Q5 (4 marks)
D4.3	Modified Euler method	2024 Q5, 2025 Q5 (4 marks)
D6.2	Expanding $f[x_0, x_1, x_2, x_3]$	2023 Q8, 2025 Q8 (4 marks)
D7.2	$f[x_0, x_1, x_2] = \Delta^2 y_0/(2!\,h^2)$	2024 Q8 (4 marks)
D8.1	Showing $[a, b]$ is a bracketing	Q9a every year (2 marks)
D9.1	Newton's method from the tangent line	2025 Q10 (3 marks)
D9.3	Iterations for $\sqrt{a}$, $\sqrt[3]{a}$ (Newton or secant)	Q11 2024, 2025 (4 marks)
D10.3	Central difference for f'' with error term	2023 Q11a (4 marks)
D10.5	Effect of h (truncation vs roundoff)	2023 Q11b (2 marks)
D11.1, D11.2	Trapezoidal rule (one strip, then composite)	2023 Q13 (5 marks)
D11.3	Simpson's rule (one pair of strips)	2024 Q13 (5 marks)
D12.3	How LU solves $A\mathbf{x} = \mathbf{b}$ (in words)	2023 Q15b (3 marks)
D12.4	Doolittle LU formulas	Q15 every year (2–3 marks)
D13.1	Positive definite by completing the square	2025 Q16a (3 marks)
D13.2	Cholesky formulas	Q16 every year (2–3 marks)

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Toolkit: five tools used again and again

Most derivations in this unit are built from the same five tools. Learn these once, and the derivations become much shorter.

T1 Taylor's theorem written with a step h

Priority: ★★ used in Weeks 3, 4, 9 and 10

Goal: a way to write $f(x_0+h)$ and $f(x_0-h)$ in terms of f and its derivatives at x_0.

Step 1. Start from Taylor's theorem (Week 2) centred on x_0:

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n + \frac{f^{(n+1)}(c)}{(n+1)!}(x-x_0)^{n+1}$$

Why: this is exact, with c somewhere between x_0 and x.

Step 2. Put $x = x_0 + h$. Then $x - x_0 = h$, so every bracket becomes a power of h:

$$f(x_0+h) = f(x_0) + h\,f'(x_0) + \frac{h^2}{2}f''(x_0) + \frac{h^3}{6}f'''(x_0) + \cdots + \frac{h^{n+1}}{(n+1)!}f^{(n+1)}(c)$$

Step 3. Replace h by $-h$. Even powers keep their sign and odd powers flip:

$$f(x_0-h) = f(x_0) - h\,f'(x_0) + \frac{h^2}{2}f''(x_0) - \frac{h^3}{6}f'''(x_0) + \cdots$$

Result. $f(x_0 \pm h) = f(x_0) \pm hf'(x_0) + \frac{h^2}{2}f''(x_0) \pm \frac{h^3}{6}f'''(x_0) + \cdots$ (signs alternate on odd powers).

T2 What $O(h^p)$ means, and the rules for using it

Priority: ★★ used in every error analysis

Goal: understand why an unknown term like $\frac{y''(c)}{2}h^2$ can be written as $O(h^2)$.

Step 1. A term is $O(h^p)$ if it is at most (some fixed constant) $\times\, h^p$ for small h.
Why: in $\frac{y''(c)}{2}h^2$ we never know c, but $|y''| \le M$ on the interval, so the term is at most $\frac{M}{2}h^2$. We keep only the part that matters: how it scales with h.

Step 2. The rules you need (each follows from "constant $\times\, h^p$"):

$h \cdot O(h^p) = O(h^{p+1})$	(multiplying by h adds one power)
$\frac{1}{h} \cdot O(h^p) = O(h^{p-1})$	(dividing by h removes one power)
$O(h^p) + O(h^p) = O(h^p)$	(two such terms is still constant $\times h^p$)
$O(h^p) + O(h^q) = O(h^p)$ if $p < q$	(for small h the lower power is bigger and dominates)

Step 3. What it predicts: if the error is $\approx Ch^p$, halving h divides the error by 2^p.

Result. $O(h)$: halve h, error $\div 2$. $O(h^2)$: error $\div 4$. $O(h^4)$: error $\div 16$.

T3 The IVT averaging trick (merging two unknown c's into one)

Priority: ★★ used in D10.2, D10.3

Goal: if g is continuous and c_1, c_2 are two points, show there is a single c between them with $g(c) = \frac{g(c_1)+g(c_2)}{2}$.

Step 1. The average of two numbers always lies between them:

$$\min\{g(c_1), g(c_2)\} \le \frac{g(c_1)+g(c_2)}{2} \le \max\{g(c_1), g(c_2)\}$$

Step 2. The Intermediate Value Theorem says a continuous function takes every value between $g(c_1)$ and $g(c_2)$ somewhere between c_1 and c_2.

Step 3. So it takes the average value at some point c between c_1 and c_2.

Result. $\dfrac{g(c_1)+g(c_2)}{2} = g(c)$ for some c between c_1 and c_2.

Why: finite-difference derivations produce two error terms with different unknown points. This trick tidies them into one.

T4 The substitution $x = x_0 + sh$ for equally spaced points

Priority: ★★ used in D7.3, D11.1–D11.3

Goal: rewrite expressions in x in terms of the "step counter" s.

Step 1. Equally spaced nodes satisfy $x_i = x_0 + ih$. Define s by $x = x_0 + sh$, i.e. $s = \frac{x-x_0}{h}$.
Why: s counts how many steps of size h you are from x_0: $s = 0$ at x_0, $s = 1$ at x_1, $s = 2$ at x_2, and so on.

Step 2. Each factor $x - x_i$ becomes a multiple of h:

$$x - x_i = (x_0 + sh) - (x_0 + ih) = (s - i)\,h$$

Step 3. For integrals: $x = x_0 + sh$ gives $\frac{dx}{ds} = h$, so $dx = h\,ds$, and the limits change from x-values to s-values:

$$\int_{x_0}^{x_1} (\cdots)\,dx = \int_0^1 (\cdots)\,h\,ds, \qquad \int_{x_0}^{x_2} (\cdots)\,dx = \int_0^2 (\cdots)\,h\,ds$$

Result. $x - x_i = (s-i)h, \quad dx = h\,ds, \quad x_i \leftrightarrow s = i.$

T5 Algebra identities that keep appearing

Priority: ★★

$$p^2 - q^2 = (p-q)(p+q) \qquad \text{(cancellation fixes; secant for } \sqrt{a})$$
$$p^3 - q^3 = (p-q)(p^2+pq+q^2) \qquad \text{(secant for } \sqrt[3]{a})$$
$$1 + r + r^2 + \cdots + r^{k-1} = \frac{1-r^k}{1-r} \quad (r \neq 1) \qquad \text{(largest floating point number)}$$
$$ax^2 + 2bxy = a\Big(x + \frac{b}{a}y\Big)^2 - \frac{b^2}{a}y^2 \qquad \text{(completing the square, positive definiteness)}$$

Check: expand the right-hand side of each line and you get the left-hand side.

Weeks 1–2: Floating point, cancellation and Taylor polynomials

D1.1 Smallest positive number $S = \beta^m$

Priority: ★★ the reasoning behind 2023 Q1b and 2025 Q1a

Goal: show that the smallest positive number in $F(\beta, k, m, M)$ is $S = \beta^m$.

Idea. a floating point number is (significand) × (base)^{exponent}. Make *both* parts as small as the rules allow.

Step 1. Every nonzero number in the system has the form

$$\pm\, d_1.d_2d_3\cdots d_k \times \beta^e, \qquad d_1 \neq 0, \qquad m \leq e \leq M$$

Why: $d_1 \neq 0$ is the "normalised" rule: it makes every number's representation unique.

Step 2. Make the significand as small as possible. The first digit cannot be 0, so the smallest choice is $d_1 = 1$. Every other digit can be 0:

$$d_1.d_2\cdots d_k = 1.00\cdots 0 = 1$$

Step 3. Make the exponent as small as possible: $e = m$, giving the factor β^m.

Step 4. Multiply the two smallest parts:

$$S = 1 \times \beta^m = \beta^m$$

Result. $S = \beta^m$. Example: $F(2, 5, -4, 4)$ gives $S = 2^{-4} = 0.0625$.

Why: any positive number smaller than S would need an exponent below m, which isn't allowed. It **underflows** to 0.

D1.2 Largest number $L = (\beta - \beta^{1-k})\,\beta^M$

Priority: ★★

Goal: show that the largest number in $F(\beta, k, m, M)$ is $L = (\beta - \beta^{1-k})\beta^M$.

Idea. make every digit as large as possible $(\beta - 1)$ and use the largest exponent M, then tidy up the significand with a geometric series.

Step 1. The largest digit in base β is $\beta - 1$ (e.g. 9 in base 10, 1 in base 2). Use it for all k digits, with exponent M:

$$L = (\beta-1).(\beta-1)\cdots(\beta-1) \times \beta^M \qquad (k \text{ digits})$$

Step 2. Write the significand as a sum of place values. The digits sit in the places $\beta^0, \beta^{-1}, \ldots, \beta^{-(k-1)}$:

$$(\beta-1)\left(1 + \beta^{-1} + \beta^{-2} + \cdots + \beta^{-(k-1)}\right)$$

Step 3. The bracket is a geometric series with k terms and ratio $r = \beta^{-1}$. Use $1 + r + \cdots + r^{k-1} = \frac{1-r^k}{1-r}$ (Toolkit T5):

$$1 + \beta^{-1} + \cdots + \beta^{-(k-1)} = \frac{1-\beta^{-k}}{1-\beta^{-1}} = \frac{\beta\,(1-\beta^{-k})}{\beta - 1}$$

Why: in the last step, multiply the top and bottom by β, so the bottom becomes $\beta - 1$.

Step 4. Multiply by $(\beta - 1)$. The $(\beta - 1)$ cancels:

$$(\beta - 1) \cdot \frac{\beta\,(1 - \beta^{-k})}{\beta - 1} = \beta(1 - \beta^{-k}) = \beta - \beta^{1-k}$$

Step 5. Attach the exponent:

$$L = (\beta - \beta^{1-k})\,\beta^{M}$$

Result. $L = (\beta - \beta^{1-k})\beta^{M}$.

Why: quick intuition: just as $9.99 = 10 - 0.01$, the largest significand is β minus one unit in the last place, and that unit is β^{1-k}.
Check: $F(2, 4, -2, 3)$: $(2 - 2^{-3}) \times 2^{3} = 1.875 \times 8 = 15$ ✓. IEEE double $F(2, 53, -1022, 1023)$: $(2 - 2^{-52}) \times 2^{1023} \approx 1.80 \times 10^{308}$ ✓

D1.3 Why rounding error is at most $u = \frac{1}{2}\beta^{1-k}$

Priority: ★ understanding (the lectures state it without proof)

Goal: show that $\dfrac{|x - fl(x)|}{|x|} \leq \frac{1}{2}\beta^{1-k}$ for any x inside the range of F.

Idea. rounding to the nearest float is wrong by at most *half a gap*. Measure that gap relative to the size of x.

Step 1. Write x in normalised form with exponent e: $x = d_1.d_2d_3\cdots \times \beta^{e}$, where the significand is between 1 and β.

Step 2. The floats with exponent e differ only in their last (kth) digit, so neighbouring floats are one unit in the last place apart. In the significand that unit is β^{1-k}, so the actual gap is

$$\text{gap} = \beta^{1-k}\,\beta^{e}$$

Step 3. Rounding to the *nearest* float means x is at most half a gap away from $fl(x)$:

$$|x - fl(x)| \leq \tfrac{1}{2}\,\beta^{1-k}\,\beta^{e}$$

Step 4. The significand is at least 1, so $|x| \geq 1 \times \beta^{e} = \beta^{e}$.

Step 5. Divide the bound in Step 3 by $|x|$, using Step 4 (a smaller denominator can only make the right-hand side bigger):

$$\frac{|x - fl(x)|}{|x|} \leq \frac{\frac{1}{2}\beta^{1-k}\beta^{e}}{\beta^{e}} = \frac{1}{2}\beta^{1-k} = u$$

Result. $\dfrac{|x - fl(x)|}{|x|} \leq u = \frac{1}{2}\beta^{1-k}$, as long as x doesn't overflow or underflow.

D2.1 Removing catastrophic cancellation from the quadratic formula

Priority: ★★★ 2025 Q1b (3 marks)

Goal: rewrite $x = \dfrac{-b + \sqrt{b^2 - 4ac}}{2a}$ so it has no catastrophic cancellation when $b^2 \gg 4ac$.

Idea. multiply by the "conjugate over itself" (which equals 1) so the subtraction turns into an addition.

Step 1. [1] **Recognise the danger.** If $b^2 \gg 4ac$, then $b^2 - 4ac \approx b^2$, so $\sqrt{b^2-4ac} \approx |b|$. For $b > 0$ the numerator is approximately $-b + b$: two nearly-equal numbers are subtracted, which is catastrophic cancellation.

Step 2. [1] **Multiply by the conjugate over itself.** Write $D = b^2 - 4ac$ to keep things short:

$$x = \frac{-b + \sqrt{D}}{2a} \times \frac{-b - \sqrt{D}}{-b - \sqrt{D}}$$

Why: the second fraction equals 1, so the value of x doesn't change, but the form does.

Step 3. **Numerator: difference of squares.** With $p = -b$ and $q = \sqrt{D}$, $(p+q)(p-q) = p^2 - q^2$:

$$(-b + \sqrt{D})(-b - \sqrt{D}) = (-b)^2 - (\sqrt{D})^2 = b^2 - (b^2 - 4ac) = 4ac$$

Why: the b^2 terms cancel *exactly in the algebra*. Exact algebraic cancellation is harmless; only cancellation of rounded numbers is dangerous.

Step 4. **Denominator:** $2a(-b - \sqrt{D}) = -2a\,(b + \sqrt{D})$.

Step 5. [1] **Simplify:**

$$x = \frac{4ac}{-2a\,(b + \sqrt{D})} = \frac{-2c}{b + \sqrt{b^2 - 4ac}}$$

Result. $x = \dfrac{-2c}{b + \sqrt{b^2 - 4ac}}$. The denominator **adds** two positive numbers, so there is no cancellation.

Why: if instead $b < 0$, it's the *other* root $\frac{-b-\sqrt{D}}{2a}$ that is dangerous. The same trick (multiply by $\frac{-b+\sqrt{D}}{-b+\sqrt{D}}$) gives $\dfrac{-2c}{b - \sqrt{D}}$.

D2.2 Other cancellation fixes

Priority: ★★ lecture problems; likely exam variations

Goal: rewrite each expression so it avoids cancellation in the case shown.

Step 1. $1 - \cos x$ for $x \approx 0$ (since $\cos x \approx 1$). Multiply by $\frac{1+\cos x}{1+\cos x}$ and use $1 - \cos^2 x = \sin^2 x$:

$$1 - \cos x = \frac{(1 - \cos x)(1 + \cos x)}{1 + \cos x} = \frac{1 - \cos^2 x}{1 + \cos x} = \frac{\sin^2 x}{1 + \cos x}$$

Step 2. $\log(x+1) - \log x$ for large x (two nearly-equal logs). Use the log rule $\log A - \log B = \log \frac{A}{B}$:

$$\log(x + 1) - \log x = \log \frac{x + 1}{x} = \log\left(1 + \frac{1}{x}\right)$$

Step 3. $\sqrt{x+1} - \sqrt{x}$ for large x. Use the conjugate:

$$\sqrt{x + 1} - \sqrt{x} = \frac{(x + 1) - x}{\sqrt{x + 1} + \sqrt{x}} = \frac{1}{\sqrt{x + 1} + \sqrt{x}}$$

Step 4. $\sqrt{x^2+1} - x$ for large x. Use the conjugate:

$$\sqrt{x^2 + 1} - x = \frac{(x^2 + 1) - x^2}{\sqrt{x^2 + 1} + x} = \frac{1}{\sqrt{x^2 + 1} + x}$$

Result. every fix turns "subtract two nearly-equal numbers" into "add" or "divide".

D2.3 Where the Taylor polynomial's coefficients come from

Priority: ★★

Goal: show that the polynomial matching $f, f', \ldots, f^{(n)}$ at x_0 is $P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$.

Idea. write a general polynomial in powers of $(x - x_0)$, then choose each coefficient so the derivatives match at x_0.

Step 1. Write a general degree-n polynomial in powers of $(x - x_0)$:

$$P(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + a_3(x - x_0)^3 + \cdots + a_n(x - x_0)^n$$

Why: at $x = x_0$, every bracket is 0, so only one term survives each time. That makes the matching easy.

Step 2. Match the value. Put $x = x_0$: every term except a_0 vanishes, so $P(x_0) = a_0$. We need $P(x_0) = f(x_0)$:

$$a_0 = f(x_0)$$

Step 3. Match the first derivative. Differentiate, then put $x = x_0$:

$$P'(x) = a_1 + 2a_2(x - x_0) + 3a_3(x - x_0)^2 + \cdots \quad \Rightarrow \quad P'(x_0) = a_1 = f'(x_0)$$

Step 4. Match the second derivative.

$$P''(x) = 2a_2 + 3 \cdot 2\, a_3(x - x_0) + \cdots \quad \Rightarrow \quad P''(x_0) = 2a_2 = f''(x_0) \quad \Rightarrow \quad a_2 = \frac{f''(x_0)}{2!}$$

Step 5. Match the third derivative.

$$P'''(x_0) = 3 \cdot 2 \cdot 1\, a_3 = 3!\, a_3 = f'''(x_0) \quad \Rightarrow \quad a_3 = \frac{f'''(x_0)}{3!}$$

Step 6. Spot the pattern. Differentiating $(x - x_0)^k$ exactly k times leaves $k!$, so in general

$$a_k = \frac{f^{(k)}(x_0)}{k!}$$

Result. $P_n(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$.

Why: the remainder $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}$ is *Taylor's theorem*. MXB103 states it without proof. It looks like "the next term", except that the derivative is evaluated at an unknown c between x_0 and x.

D2.4 Taylor polynomial and error bound: the full exam answer

Priority: ★★★ Q2 every year (6 marks)

Goal: (2025 Q2) (a) find P_2 for $f(x) = \cos x$ centred on $x_0 = 0$; (b) bound the error for $0 \le x \le \pi/3$.

Idea. (a) differentiate, evaluate at x_0, substitute. (b) write the remainder, then replace each unknown by its worst case.

Part (a)

Step 1. [1] Write the form of the Taylor polynomial:

$$P_2(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2$$

Step 2. [1] Differentiate and evaluate at $x_0 = 0$:

$$f(x) = \cos x \Rightarrow f(0) = 1, \qquad f'(x) = -\sin x \Rightarrow f'(0) = 0, \qquad f''(x) = -\cos x \Rightarrow f''(0) = -1$$

Step 3. [1] Substitute:

$$P_2(x) = 1 + 0 \cdot x + \frac{-1}{2}x^2 = 1 - \frac{x^2}{2}$$

Part (b)

Step 4. [1] Write the remainder, and say where c is. The next derivative is $f'''(x) = \sin x$:

$$R_2(x) = \frac{f'''(c)}{3!}(x - 0)^3 = \frac{\sin c}{6}\, x^3, \qquad c \text{ between } 0 \text{ and } x$$

Why: x is between 0 and $\pi/3$, so c is too.

Step 5. [1] Bound the derivative factor. On $[0, \pi/3]$, sin is increasing, so its largest value is at the right end:

$$|\sin c| \le \sin \tfrac{\pi}{3} = \tfrac{\sqrt{3}}{2}$$

Step 6. Bound the distance factor. The farthest point of $[0, \pi/3]$ from $x_0 = 0$ is $\pi/3$:

$$|x|^3 \le \left(\frac{\pi}{3}\right)^3$$

Step 7. [1] Multiply:

$$|f(x) - P_2(x)| = |R_2(x)| \le \frac{1}{6} \cdot \frac{\sqrt{3}}{2} \cdot \left(\frac{\pi}{3}\right)^3 = 0.1658$$

Result. (a) $P_2(x) = 1 - \dfrac{x^2}{2}$. (b) Absolute error ≤ 0.1658 for all $0 \le x \le \pi/3$.

Common mistake: when $x_0 \neq 0$, the distance factor is the farthest distance *from* x_0, not the right endpoint. In 2023 Q2 ($x_0 = \pi/6$ on $[0, \pi/3]$) it is $\max(|0 - \frac{\pi}{6}|, |\frac{\pi}{3} - \frac{\pi}{6}|) = \frac{\pi}{6}$, giving the bound $\frac{1}{2}\left(\frac{\pi}{6}\right)^2 = \frac{\pi^2}{72}$.

Weeks 3–4: ODE methods

Setting for every derivation in this section. We solve $\frac{dy}{dt} = f(t, y)$ on $a \le t \le b$ with $y(a) = \alpha$. We use the grid $h = \frac{b-a}{n}$, $t_i = a + ih$, so $t_{i+1} - t_i = h$. We write $y_i = y(t_i)$ for the exact (unknown) values and $w_i \approx y_i$ for the computed ones.

The same four moves appear in every ODE derivation:

1	Taylor-expand $y(t)$ about t_i
2	Evaluate at t_{i+1}, so every $(t - t_i)$ becomes h
3	Use the ODE to replace derivatives of y by f
4	Drop the $O(h^{\cdot})$ term and replace y_i by w_i

D3.1 Euler's method

Priority: ★★ the template for every ODE derivation

Goal: derive $w_{i+1} = w_i + h\, f(t_i, w_i)$.

Idea. a degree-1 Taylor polynomial is the tangent line. Follow the tangent for one step.

Step 1. Move 1. Taylor polynomial of degree 1 for $y(t)$ about t_i, with remainder:

$$y(t) = y(t_i) + y'(t_i)(t - t_i) + \frac{y''(c)}{2}(t - t_i)^2$$

Why: we know y at t_i and want it at the next time. Taylor predicts nearby values from information at one point.

Step 2. Move 2. Evaluate at $t = t_{i+1}$. Since $t_{i+1} - t_i = h$:

$$y_{i+1} = y_i + h\, y'(t_i) + \frac{y''(c)}{2}h^2 = y_i + h\, y'(t_i) + O(h^2)$$

Why: c is unknown but y'' is bounded, so the last term is at most a constant times h^2 (Toolkit T2).

Step 3. Move 3. The ODE says $y'(t) = f(t, y(t))$. At $t = t_i$ this gives $y'(t_i) = f(t_i, y_i)$:

$$y_{i+1} = y_i + h\, f(t_i, y_i) + O(h^2)$$

Step 4. Move 4. Drop the $O(h^2)$ term, and replace the exact y_i by the approximations w_i:

$$w_{i+1} = w_i + h\, f(t_i, w_i), \qquad w_0 = \alpha$$

Result. Euler's method: $w_{i+1} = w_i + h\, f(t_i, w_i)$.

Why: the term we dropped, $O(h^2)$, is exactly the error made in one step: the **local error**.

D3.2 Local error $O(h^{p+1})$ gives global error $O(h^p)$

Priority: ★★ explains "Euler is first order"

Goal: show why a method whose single step has error $O(h^{p+1})$ has total error $O(h^p)$.

Idea. the total error is roughly (error per step) × (number of steps), and the number of steps is $\frac{b-a}{h}$.

Step 1. Each step adds an error of at most $C\, h^{p+1}$, for some constant C (the local error).

Step 2. Getting from a to b takes $n = \dfrac{b-a}{h}$ steps.

Step 3. Add up all the steps' errors (roughly, ignoring how earlier errors are carried forward):

$$\text{global error} \approx n \times C\,h^{p+1} = \frac{b-a}{h} \times C\,h^{p+1} = C(b-a)\,h^p = O(h^p)$$

Why: one power of h is lost because a smaller h means *more* steps. A careful proof also tracks how earlier errors propagate, but the conclusion is the same.

Result. local $O(h^{p+1}) \Rightarrow$ global $O(h^p)$. Euler: local $O(h^2)$, so global $O(h)$ (**first order**).

D4.1 The total derivative $f'(t, y)$

Priority: ★★★ every Taylor 2 question: 2025 gave 0 marks if this is wrong

Goal: show $f'(t,y) = \dfrac{\partial f}{\partial t} + \dfrac{\partial f}{\partial y} \cdot f(t,y)$, and that $y'' = f'(t,y)$.

Idea. along the solution, y is itself a function of t, so t affects f in two ways: directly, and through y.

Step 1. Along the solution curve, $y = y(t)$. So $f(t, y(t))$ is really a function of t alone, but t enters it twice: in the first slot directly, and in the second slot through $y(t)$.

Step 2. The chain rule for a function of two variables adds up both effects:

$$\frac{d}{dt} f\big(t, y(t)\big) = \frac{\partial f}{\partial t} \cdot \frac{dt}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}$$

Step 3. Simplify: $\frac{dt}{dt} = 1$, and $\frac{dy}{dt} = y' = f(t,y)$ by the ODE:

$$f'(t,y) = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial y} \cdot f(t,y)$$

Step 4. Since $y' = f(t,y)$, differentiating once more gives $y'' = \frac{d}{dt} f(t,y) = f'(t,y)$.

Result. $f'(t,y) = f_t + f_y \cdot f$, and $y'' = f'(t,y)$.

Worked (lecture): $f(t,y) = 3t - \dfrac{y}{t}$.

$$\begin{aligned}
\frac{\partial f}{\partial t} &= 3 + \frac{y}{t^2} && \text{(treat } y \text{ as a constant: } \tfrac{d}{dt}(-yt^{-1}) = yt^{-2}) \\
\frac{\partial f}{\partial y} &= -\frac{1}{t} && \text{(treat } t \text{ as a constant)} \\
f'(t,y) &= 3 + \frac{y}{t^2} + \Big(-\frac{1}{t}\Big)\Big(3t - \frac{y}{t}\Big) && (f_t + f_y \cdot f) \\
&= 3 + \frac{y}{t^2} - 3 + \frac{y}{t^2} = \frac{2y}{t^2}
\end{aligned}$$

Common mistake: writing $f' = 3 + \frac{y}{t^2}$, i.e. forgetting the $f_y \cdot f$ part. In MATLAB this is the missing `*f(t,y)` bug (2023 Q4, 2024 Q4).

D4.2 Second-order Taylor method

Priority: ★★★ 2023 Q5 (4 marks)

Goal: derive $w_{i+1} = w_i + h\,f(t_i, w_i) + \dfrac{h^2}{2} f'(t_i, w_i)$.

Idea. the same four moves as Euler, but keep one more Taylor term.

Step 1. [1] **Move 1.** Taylor polynomial of degree 2 for $y(t)$ about t_i:

$$y(t) = y(t_i) + y'(t_i)(t - t_i) + \frac{y''(t_i)}{2}(t - t_i)^2 + O\big((t - t_i)^3\big)$$

Step 2. [1] **Move 2.** Evaluate at $t = t_{i+1}$, so $t_{i+1} - t_i = h$:

$$y_{i+1} = y_i + h\,y'(t_i) + \frac{h^2}{2} y''(t_i) + O(h^3)$$

Step 3. [1] **Move 3.** Replace the derivatives of y using the ODE and D4.1: $y' = f(t, y)$ and $y'' = f'(t, y)$:

$$y_{i+1} = y_i + h\,f(t_i, y_i) + \frac{h^2}{2} f'(t_i, y_i) + O(h^3)$$

Step 4. [1] **Move 4.** Drop $O(h^3)$, and replace y_i with w_i:

$$w_{i+1} = w_i + h\,f(t_i, w_i) + \frac{h^2}{2} f'(t_i, w_i)$$

Result. second-order Taylor method. Local error $O(h^3)$, so global $O(h^2)$ (second order).

D4.3 Modified Euler method

Priority: ★★★ 2024 Q5 and 2025 Q5 (4 marks each)

Goal: derive $w_{i+1} = w_i + \dfrac{h}{2}\Big[f(t_i, w_i) + f\big(t_{i+1},\, w_i + hf(t_i, w_i)\big)\Big]$.

Idea. start from Taylor 2, but get rid of the awkward f' by approximating it with a difference quotient (a slope). Then patch the one remaining problem using Euler.

Step 1. [0.5] **Taylor polynomial of degree 2** for $y(t)$ about t_i:

$$y(t) = y(t_i) + y'(t_i)(t - t_i) + \frac{y''(t_i)}{2}(t - t_i)^2 + O\big((t - t_i)^3\big)$$

Step 2. [0.5] **Evaluate at** t_{i+1}, so $t_{i+1} - t_i = h$:

$$y_{i+1} = y_i + h\,y'(t_i) + \frac{h^2}{2} y''(t_i) + O(h^3)$$

Step 3. [0.5] **Replace the derivatives of** y **by** f ($y' = f$, $y'' = f'$):

$$y_{i+1} = y_i + h\,f(t_i, y_i) + \frac{h^2}{2} f'(t_i, y_i) + O(h^3)$$

Why: up to here this is just Taylor 2. Next we remove f', because computing it needs the chain rule.

Step 4. [0.5] **Approximate** f' **by a forward difference**, the slope of f between t_i and t_{i+1} (Week 10, D10.1):

$$f'(t_i, y_i) = \frac{f(t_{i+1}, y_{i+1}) - f(t_i, y_i)}{h} + O(h)$$

Step 5. [1] **Substitute and simplify.** To keep it readable, write $f_i = f(t_i, y_i)$ and $f_{i+1} = f(t_{i+1}, y_{i+1})$:

$$\begin{aligned} y_{i+1} &= y_i + hf_i + \frac{h^2}{2}\left[\frac{f_{i+1} - f_i}{h} + O(h)\right] + O(h^3) \\ &= y_i + hf_i + \frac{h}{2}(f_{i+1} - f_i) + O(h^3) && (\tfrac{h^2}{2} \cdot \tfrac{1}{h} = \tfrac{h}{2}, \text{ and } \tfrac{h^2}{2} \cdot O(h) = O(h^3)) \\ &= y_i + \frac{h}{2}f_i + \frac{h}{2}f_{i+1} + O(h^3) && (hf_i - \tfrac{h}{2}f_i = \tfrac{h}{2}f_i) \\ &= y_i + \frac{h}{2}\big[f(t_i, y_i) + f(t_{i+1}, y_{i+1})\big] + O(h^3) \end{aligned}$$

Step 6. **Drop $O(h^3)$ and replace y by w:**

$$w_{i+1} = w_i + \frac{h}{2}\Big[f(t_i, w_i) + f(t_{i+1},\, w_{i+1})\Big]$$

Step 7. [1] **Patch the problem: w_{i+1} appears on both sides.** We are trying to *compute* w_{i+1}, so it can't be on the right-hand side. Inside f on the right, replace it with Euler's prediction $w_{i+1} \approx w_i + h\, f(t_i, w_i)$:

$$w_{i+1} = w_i + \frac{h}{2}\Big[f(t_i, w_i) + f\big(t_{i+1},\, w_i + h\, f(t_i, w_i)\big)\Big]$$

Why: Euler's error inside f is $O(h^2)$. It's multiplied by $\frac{h}{2}$, so it only adds $O(h^3)$, which is the same size as the error we already dropped. The method stays second order.

Step 8. **(Optional) the k-form.** Name the two increments $k_1 = h\, f(t_i, w_i)$ and $k_2 = h\, f(t_i + h,\, w_i + k_1)$. Then $\frac{h}{2}[\,\cdot\,] = \frac{1}{2}(k_1 + k_2)$:

$$w_{i+1} = w_i + \tfrac{1}{2}(k_1 + k_2)$$

Result. Modified Euler: $w_{i+1} = w_i + \frac{h}{2}\big[f(t_i, w_i) + f(t_{i+1}, w_i + hf(t_i, w_i))\big]$. Global $O(h^2)$, with no f' needed.

D4.4 RK4: what you need to know (no derivation required)

Priority: ★

Goal: know the RK4 formulas, and see why they make sense.

Step 1. The lectures state RK4 without derivation (it takes "laborious algebra"). You need the formulas and the order (global $O(h^4)$) only:

$$\begin{aligned} k_1 &= hf(t_i, w_i), & k_2 &= hf\big(t_i + \tfrac{h}{2},\, w_i + \tfrac{k_1}{2}\big), \\ k_3 &= hf\big(t_i + \tfrac{h}{2},\, w_i + \tfrac{k_2}{2}\big), & k_4 &= hf(t_i + h,\, w_i + k_3) \end{aligned}$$

$$w_{i+1} = w_i + \tfrac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Step 2. **Sanity check: if f depends only on t**, then $k_2 = k_3 = hf(t_i + \frac{h}{2})$, and

$$w_{i+1} - w_i = \frac{h}{6}\Big[f(t_i) + 4f\big(t_i + \tfrac{h}{2}\big) + f(t_{i+1})\Big]$$

Why: that is exactly Simpson's rule (D11.3) on $[t_i, t_{i+1}]$, using the strip half-width $\frac{h}{2}$: $\frac{h/2}{3}[f + 4f + f]$. So RK4 really is "Simpson's rule for ODEs", which is why the weights are $1, 2, 2, 1$.

Result. RK4 is a weighted average of four slope samples, and reduces to Simpson's rule when $f = f(t)$.

Weeks 5–7: Interpolation

Setting. We have data points $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$ with *distinct* x_i. We want the polynomial P_n of degree $\leq n$ with $P_n(x_i) = y_i$ for every i. The key theorem: **such a polynomial exists and is unique**. Every "form" below is the same polynomial, written differently.

D5.1 From two points to the Lagrange form

Priority: ★★ lecture derivation; shows where Lagrange comes from

Goal: find the line through (x_0, y_0) and (x_1, y_1), and rearrange it into $P_1(x) = \dfrac{x - x_1}{x_0 - x_1} y_0 + \dfrac{x - x_0}{x_1 - x_0} y_1$.

Idea. solve for the line directly, then regroup the result by y_0 and y_1.

Step 1. Look for $P_1(x) = a_0 + a_1 x$. It must pass through both points:

$$a_0 + a_1 x_0 = y_0 \quad (1) \qquad\qquad a_0 + a_1 x_1 = y_1 \quad (2)$$

Step 2. Subtract (1) from (2) to eliminate a_0:

$$a_1(x_1 - x_0) = y_1 - y_0 \quad \Rightarrow \quad a_1 = \frac{y_1 - y_0}{x_1 - x_0} \quad \text{(the slope)}$$

Step 3. Substitute back into (1) and solve for a_0:

$$a_0 = y_0 - a_1 x_0 = \frac{y_0(x_1 - x_0) - (y_1 - y_0)x_0}{x_1 - x_0} = \frac{x_1 y_0 - x_0 y_1}{x_1 - x_0}$$

Step 4. Put both into P_1 over a common denominator, then collect the y_0 terms and the y_1 terms:

$$P_1(x) = \frac{x_1 y_0 - x_0 y_1 + (y_1 - y_0)x}{x_1 - x_0} = \frac{y_0\,(x_1 - x) + y_1\,(x - x_0)}{x_1 - x_0}$$

Step 5. Split into two fractions. For the y_0 term, flip the signs of the top and bottom ($\frac{x_1 - x}{x_1 - x_0} = \frac{x - x_1}{x_0 - x_1}$):

$$P_1(x) = \underbrace{\frac{x - x_1}{x_0 - x_1}}_{L_{1,0}(x)} y_0 + \underbrace{\frac{x - x_0}{x_1 - x_0}}_{L_{1,1}(x)} y_1$$

Result. $P_1(x) = L_{1,0}(x)\, y_0 + L_{1,1}(x)\, y_1$. Note that $L_{1,0}$ equals 1 at x_0 and 0 at x_1, and $L_{1,1}$ does the opposite. That pattern is what generalises.

D5.2 Why the Lagrange form interpolates

Priority: ★★

Goal: show that $P_n(x) = \sum_{i=0}^{n} L_{n,i}(x)\, y_i$ with $L_{n,i}(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$ passes through every data point.

Idea. each $L_{n,i}$ is designed to be 1 at its own node and 0 at all the others.

Step 1. $L_{n,i}$ **is zero at the other nodes.** Its numerator $\prod_{j \neq i}(x - x_j)$ contains the factor

$(x - x_k)$ for every $k \neq i$. So at $x = x_k$ that factor is 0:

$$L_{n,i}(x_k) = 0 \qquad (k \neq i)$$

Step 2. $L_{n,i}$ **is one at its own node.** At $x = x_i$ the numerator becomes $\prod_{j \neq i}(x_i - x_j)$, which is exactly the denominator:

$$L_{n,i}(x_i) = \frac{\prod_{j \neq i}(x_i - x_j)}{\prod_{j \neq i}(x_i - x_j)} = 1$$

Why: this is why the nodes must be distinct: if $x_i = x_j$, the denominator would be 0.

Step 3. Evaluate P_n at a node x_k. Every term with $i \neq k$ vanishes, and the $i = k$ term has $L_{n,k}(x_k) = 1$:

$$P_n(x_k) = \sum_i L_{n,i}(x_k)\, y_i = L_{n,k}(x_k)\, y_k = y_k$$

Step 4. Degree. Each $L_{n,i}$ is a product of n linear factors, so it has degree n. A sum of these has degree at most n.

Result. P_n has degree $\leq n$ and $P_n(x_k) = y_k$ for every k, so it is *the* interpolating polynomial.

Written out for three points:

$$P_2(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} y_0 + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} y_1 + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} y_2$$

D5.3 Why the Lagrange basis polynomials add up to 1

Priority: ★ the arithmetic check

Goal: show $\sum_i L_{n,i}(x) = 1$ for every x.

Step 1. Interpolate the constant function $f(x) = 1$. Then every $y_i = 1$, so the Lagrange form gives $P_n(x) = \sum_i L_{n,i}(x) \cdot 1$.

Step 2. The constant polynomial 1 also passes through all these points, and it has degree $0 \leq n$.

Step 3. By uniqueness, these are the same polynomial: $\sum_i L_{n,i}(x) = 1$.

Result. $\sum_i L_{n,i}(x) = 1$. If your computed L values don't add to 1, one of them is wrong.

D5.4 Using the interpolation remainder (two cases)

Priority: ★★ 2024 Q6b (3 marks)

Goal: bound $|f(x) - P_n(x)|$ using $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)(x - x_1)\cdots(x - x_n)$, which is stated without proof.

Step 1. Write R_n, with c in the interval containing the nodes and x.

Step 2. Bound $|f^{(n+1)}(c)|$ by its **maximum** over that interval (c is unknown, so use the worst case).

Step 3. Handle the product:

- error at **one point** $\bar{x}$: just *evaluate* $|(\bar{x} - x_0)\cdots(\bar{x} - x_n)|$;
- error for **all** x **in an interval**: use the *maximum* of the product (usually given in the exam).

Step 4. Multiply and divide by $(n+1)!$.
Worked (2024 Q6b): $f = \cos x$, nodes $0, \frac{\pi}{6}, \frac{\pi}{2}$, all $x \in [0, \frac{\pi}{4}]$, given $|x(x-\frac{\pi}{6})(x-\frac{\pi}{2})| < 0.17$:

$$|R_2(x)| = \left|\frac{\sin c}{3!}\right| \left|x\left(x-\frac{\pi}{6}\right)\left(x-\frac{\pi}{2}\right)\right| < \frac{1}{6} \times 0.17 = 0.0283$$

Why: $f'''(x) = \sin x$, and $|\sin c| \le 1$.

Result. error $\le \dfrac{\max|f^{(n+1)}|}{(n+1)!} \times |\text{product}|$ (evaluated at one point, or its max over the interval).

D6.1 Newton divided difference coefficients

Priority: ★★ lecture derivation

Goal: show that the coefficients of $P_n(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + \cdots$ are the divided differences $a_k = f[x_0, \ldots, x_k]$, by checking a_0, a_1 and a_2.

Idea. impose $P(x_0) = y_0$, $P(x_1) = y_1$, $P(x_2) = y_2$ one at a time. Each new condition brings in only one new coefficient.

Step 1. At x_0: every term except a_0 contains $(x - x_0)$, so they vanish:

$$a_0 = y_0 = f[x_0]$$

Step 2. At x_1: terms from a_2 onwards contain $(x - x_1)$, so they vanish:

$$a_0 + a_1(x_1 - x_0) = y_1 \quad \Rightarrow \quad a_1 = \frac{y_1 - y_0}{x_1 - x_0} = f[x_0, x_1]$$

Step 3. At x_2: terms from a_3 onwards vanish. Solve for a_2:

$$a_2 = \frac{y_2 - y_0 - a_1(x_2 - x_0)}{(x_2 - x_0)(x_2 - x_1)}$$

Step 4. Simplify the numerator. Split $y_2 - y_0 = (y_2 - y_1) + (y_1 - y_0)$ and use $a_1 = \frac{y_1 - y_0}{x_1 - x_0}$:

$$\begin{aligned} y_2 - y_0 - a_1(x_2 - x_0) &= (y_2 - y_1) + (y_1 - y_0)\left[1 - \frac{x_2 - x_0}{x_1 - x_0}\right] \\ &= (y_2 - y_1) + (y_1 - y_0) \cdot \frac{x_1 - x_2}{x_1 - x_0} = (y_2 - y_1) - (y_1 - y_0)\frac{x_2 - x_1}{x_1 - x_0} \end{aligned}$$

Step 5. Divide by $(x_2 - x_0)(x_2 - x_1)$. Dividing the first piece by $(x_2 - x_1)$ gives a slope, and the $(x_2 - x_1)$ cancels in the second:

$$a_2 = \frac{\dfrac{y_2 - y_1}{x_2 - x_1} - \dfrac{y_1 - y_0}{x_1 - x_0}}{x_2 - x_0} = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = f[x_0, x_1, x_2]$$

Result. $a_0 = f[x_0]$, $a_1 = f[x_0, x_1]$, $a_2 = f[x_0, x_1, x_2]$, and in general $a_k = f[x_0, \ldots, x_k]$, the top diagonal of the table.

D6.2 Expanding $f[x_0, x_1, x_2, x_3]$

Priority: ★★★ 2023 Q8 and 2025 Q8 (4 marks, 1 per line)

Goal: express $f[x_0, x_1, x_2, x_3]$ using only the x_i and y_i.

Idea. apply the definition (drop the first point) − (drop the last point), all over (last x − first x), again and again until only $f[x_j] = y_j$ remain.

Step 1. [1] Apply the definition to the whole thing:

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0}$$

Step 2. [1] Apply it to each second divided difference:

$$= \frac{\dfrac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1} - \dfrac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}}{x_3 - x_0}$$

Step 3. [1] Apply it to each first divided difference, $f[x_j, x_{j+1}] = \dfrac{f[x_{j+1}] - f[x_j]}{x_{j+1} - x_j}$:

$$= \frac{\dfrac{\frac{f[x_3]-f[x_2]}{x_3-x_2} - \frac{f[x_2]-f[x_1]}{x_2-x_1}}{x_3 - x_1} - \dfrac{\frac{f[x_2]-f[x_1]}{x_2-x_1} - \frac{f[x_1]-f[x_0]}{x_1-x_0}}{x_2 - x_0}}{x_3 - x_0}$$

Step 4. [1] Replace each $f[x_j]$ by y_j:

$$f[x_0, x_1, x_2, x_3] = \frac{\dfrac{\frac{y_3-y_2}{x_3-x_2} - \frac{y_2-y_1}{x_2-x_1}}{x_3 - x_1} - \dfrac{\frac{y_2-y_1}{x_2-x_1} - \frac{y_1-y_0}{x_1-x_0}}{x_2 - x_0}}{x_3 - x_0}$$

Result. the 4-line expansion above. Write every level on its own line: each one is a mark.

Common mistake: dividing by the neighbours' gap (e.g. $x_3 - x_2$) at the second level. A second divided difference over x_1, x_2, x_3 divides by the *outer* gap $x_3 - x_1$.

D7.1 Second and third forward differences

Priority: ★★

Goal: express $\Delta^2 y_i$ and $\Delta^3 y_i$ in terms of the y's.

Idea. Δ means "next minus current". Apply it repeatedly.

Step 1. Definition: $\Delta y_i = y_{i+1} - y_i$.

Step 2. Second difference = Δ applied to Δy_i:

$$\Delta^2 y_i = \Delta y_{i+1} - \Delta y_i = (y_{i+2} - y_{i+1}) - (y_{i+1} - y_i) = y_{i+2} - 2y_{i+1} + y_i$$

Step 3. Third difference = Δ applied to $\Delta^2 y_i$:

$$\Delta^3 y_i = \Delta^2 y_{i+1} - \Delta^2 y_i = (y_{i+3} - 2y_{i+2} + y_{i+1}) - (y_{i+2} - 2y_{i+1} + y_i) = y_{i+3} - 3y_{i+2} + 3y_{i+1} - y_i$$

Result. the coefficients are Pascal's triangle with alternating signs: $1, -2, 1$ and $1, -3, 3, -1$.

D7.2 Divided differences equal scaled forward differences

Priority: ★★★ 2024 Q8 (4 marks)

Goal: for equally spaced nodes (spacing h), show $f[x_0, x_1, x_2] = \dfrac{\Delta^2 y_0}{2!\, h^2}$.

Idea. write out the divided difference, then use equal spacing: every gap between neighbours is h, and the outer gap is $2h$.

Step 1. Warm-up, $k = 1$. $f[x_0, x_1] = \dfrac{y_1 - y_0}{x_1 - x_0} = \dfrac{\Delta y_0}{h} = \dfrac{\Delta y_0}{1!\, h^1}$ ✓

Step 2. [1] Apply the definition:

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$$

Step 3. [1] Write out the first divided differences:

$$= \frac{\dfrac{y_2 - y_1}{x_2 - x_1} - \dfrac{y_1 - y_0}{x_1 - x_0}}{x_2 - x_0}$$

Step 4. Use equal spacing: $x_2 - x_1 = x_1 - x_0 = h$ and $x_2 - x_0 = 2h$:

$$= \frac{\dfrac{y_2 - y_1}{h} - \dfrac{y_1 - y_0}{h}}{2h}$$

Step 5. [2] Simplify. Multiply the top and bottom by h, then recognise $\Delta^2 y_0$ (D7.1):

$$= \frac{(y_2 - y_1) - (y_1 - y_0)}{2h^2} = \frac{y_2 - 2y_1 + y_0}{2h^2} = \frac{\Delta^2 y_0}{2!\, h^2}$$

Result. $f[x_0, x_1, x_2] = \dfrac{\Delta^2 y_0}{2!\, h^2}$. In general $f[x_0, \ldots, x_k] = \dfrac{\Delta^k y_0}{k!\, h^k}$.

D7.3 The Newton forward difference form

Priority: ★★

Goal: for equally spaced nodes, turn the divided-difference form into $P_n(x) = \displaystyle\sum_{k=0}^{n} \binom{s}{k} \Delta^k y_0$ with $s = \frac{x - x_0}{h}$.

Idea. substitute $x = x_0 + sh$ (Toolkit T4). Every power of h cancels.

Step 1. The divided difference form (D6.1): term k is $f[x_0, \ldots, x_k]\,(x - x_0)(x - x_1)\cdots(x - x_{k-1})$.

Step 2. With $x = x_0 + sh$, each factor is $x - x_i = (s - i)h$, so the product of the first k factors is

$$(x - x_0)(x - x_1)\cdots(x - x_{k-1}) = h^k\, s(s-1)(s-2)\cdots(s-k+1)$$

Step 3. The coefficient, from D7.2: $f[x_0, \ldots, x_k] = \dfrac{\Delta^k y_0}{k!\, h^k}$.

Step 4. Multiply them. The h^k cancels:

$$\frac{\Delta^k y_0}{k!\,h^k}\cdot h^k\, s(s-1)\cdots(s-k+1) = \frac{s(s-1)\cdots(s-k+1)}{k!}\,\Delta^k y_0 = \binom{s}{k}\Delta^k y_0$$

Result. $P_n(x) = y_0 + s\,\Delta y_0 + \dfrac{s(s-1)}{2!}\Delta^2 y_0 + \dfrac{s(s-1)(s-2)}{3!}\Delta^3 y_0 + \cdots, \qquad s = \dfrac{x - x_0}{h}.$

Weeks 8–9: Root finding

Setting. We want x with $f(x) = 0$. Any equation can be put in this form by moving everything to one side.

D8.1 Showing $[a, b]$ is a bracketing

Priority: ★★★ Q9a every year (2 marks)

Goal: show that f has a root in $[a, b]$.

Idea. if a continuous function is negative at one end and positive at the other, it must cross zero in between (Intermediate Value Theorem).

Step 1. [0.5 + 0.5] Evaluate f at both ends.

Step 2. [1] State: they have **opposite signs**, and f is **continuous** on $[a, b]$, so by the IVT f takes every value between $f(a)$ and $f(b)$, including 0. Hence $[a, b]$ is a bracketing.
Worked (2025 Q9a): $f(x) = x^3 - 2x - 7$ on $[1, 3]$.

$$f(1) = 1 - 2 - 7 = -8 < 0, \qquad f(3) = 27 - 6 - 7 = 14 > 0$$

Opposite signs, and f is a polynomial (so continuous). Hence $[1, 3]$ is a bracketing for f.

Result. "$f(a)$ and $f(b)$ have opposite signs and f is continuous, so $[a, b]$ brackets a root."

D8.2 Bisection error bound

Priority: ★★

Goal: show that after n iterations the midpoint p_n satisfies $|p_n - r| \le \dfrac{b-a}{2^n}$, where r is the root.

Idea. every iteration halves the bracket, and the root is always inside it.

Step 1. The starting bracket has length $b - a$, and each iteration halves it. So the bracket used in iteration n has length $\dfrac{b-a}{2^{n-1}}$.

Step 2. p_n is the midpoint of that bracket, and the root r is somewhere inside it. A point inside an interval is at most half its length from the midpoint:

$$|p_n - r| \le \frac{1}{2} \cdot \frac{b-a}{2^{n-1}} = \frac{b-a}{2^n}$$

Step 3. How many iterations for a tolerance? We need $\frac{b-a}{2^n} \le \text{tol}$, i.e. $2^n \ge \frac{b-a}{\text{tol}}$, so $n \ge \log_2 \frac{b-a}{\text{tol}}$.

Result. $|p_n - r| \le \dfrac{b-a}{2^n}$. E.g. on $[1, 3]$ with tol 10^{-3}: $2^n \ge 2000$, so $n = 11$.

D8.3 Rearranging $f(x) = 0$ into $x = g(x)$

Priority: ★★ lecture example

Goal: find two fixed point forms of $0.9 \cos x = \sqrt{x}$ on $[0, 1]$.

Idea. any algebra that isolates an x on one side gives a valid g. Different choices behave differently.

Step 1. Rearrangement 1. Divide by 0.9, then apply $\cos^{-1}$:

$$\cos x = \frac{\sqrt{x}}{0.9} \quad \Rightarrow \quad x = \cos^{-1}\left(\frac{\sqrt{x}}{0.9}\right) = g_1(x)$$

Step 2. Rearrangement 2. Square both sides (safe here, since both sides are ≥ 0 on $[0, 1]$):

$$0.81\cos^2 x = x \quad \Rightarrow \quad x = 0.81\cos^2 x = g_2(x)$$

Result. same equation, two g's. From $x_0 = 0.5$, g_1 diverges (then becomes undefined), and g_2 converges to 0.57. D8.4 explains why.

D8.4 Why $|g'| < 1$ makes fixed point iteration converge

Priority: ★ explains the fixed point theorem and linear convergence

Goal: show that if $|g'(x)| \leq k < 1$, the errors $|x_n - p|$ shrink to 0.

Idea. compare each iterate's distance to the fixed point p with the previous one, using the Mean Value Theorem.

Step 1. The fixed point satisfies $p = g(p)$. Subtract this from $x_{n+1} = g(x_n)$:

$$x_{n+1} - p = g(x_n) - g(p)$$

Step 2. Mean Value Theorem: $g(x_n) - g(p) = g'(\xi_n)\,(x_n - p)$ for some ξ_n between x_n and p:

$$x_{n+1} - p = g'(\xi_n)\,(x_n - p)$$

Step 3. Take absolute values, and use $|g'| \leq k$:

$$|x_{n+1} - p| \leq k\,|x_n - p|$$

Step 4. Apply this repeatedly:

$$|x_n - p| \leq k\,|x_{n-1} - p| \leq k^2|x_{n-2} - p| \leq \cdots \leq k^n|x_0 - p| \;\longrightarrow\; 0 \quad \text{since } 0 \leq k < 1$$

Result. the error shrinks by a factor of about $\lambda = |g'(p)|$ each step: **linear** convergence. If $|g'(p)| > 1$ the error *grows*.

D9.1 Newton's method from the tangent line

Priority: ★★★ 2025 Q10 (3 marks)

Goal: derive $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$.

Idea. replace the curve by its tangent line at x_n. Where the tangent hits zero is the next guess.

Step 1. [1] The tangent line to f at $x = x_n$ (the degree-1 Taylor polynomial):

$$f(x) \approx f(x_n) + f'(x_n)\,(x - x_n)$$

Step 2. [1] Let x_{n+1} be where the tangent line crosses the x-axis, i.e. set it equal to 0:

$$0 = f(x_n) + f'(x_n)\,(x_{n+1} - x_n)$$

Step 3. [1] Solve for x_{n+1}: subtract $f(x_n)$, divide by $f'(x_n)$, then add x_n:

$$x_{n+1} - x_n = -\frac{f(x_n)}{f'(x_n)} \quad \Rightarrow \quad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Result. Newton's method: $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$. It needs $f'(x_n) \neq 0$.

D9.2 The secant method from Newton's method

Priority: ★★

Goal: derive $x_{n+1} = x_n - f(x_n)\dfrac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$.

Idea. Newton needs $f'(x_n)$. Replace it by the slope of the line through the last two points.

Step 1. Approximate the derivative by the secant slope (rise over run between the last two iterates):

$$f'(x_n) \approx \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}$$

Step 2. Substitute into Newton. Dividing by a fraction means multiplying by its reciprocal:

$$x_{n+1} = x_n - \frac{f(x_n)}{\dfrac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}} = x_n - f(x_n)\,\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$$

Result. secant method. It needs two starting values and no derivative, and its order is about 1.618 (less than Newton's 2).

D9.3 Iterations for $\sqrt{a}$, $\sqrt[3]{a}$ and more

Priority: ★★★ Q11: 2024 (Newton, $\sqrt[3]{a}$) and 2025 (secant, $\sqrt{a}$), 4 marks

Goal: derive an iteration for a root of a number $a > 0$.

Idea. choose $f(x)$ whose root *is* the number you want (for $\sqrt[k]{a}$, use $f(x) = x^k - a$), apply the method, then simplify.

(a) Heron's method: Newton for $\sqrt{a}$ (lecture)

Step 1. Choose $f(x) = x^2 - a$ (its root is $\sqrt{a}$), so $f'(x) = 2x$.

Step 2. Newton, then split the fraction:

$$x_{n+1} = x_n - \frac{x_n^2 - a}{2x_n} = x_n - \frac{x_n}{2} + \frac{a}{2x_n} = \frac{1}{2}\left(x_n + \frac{a}{x_n}\right)$$

(b) Newton for $\sqrt[3]{a}$ (2024 Q11)

Step 3. [1.5] Choose $f(x) = x^3 - a$ (its root is $\sqrt[3]{a}$), so $f'(x) = 3x^2$.

Step 4. [1.5] Apply Newton:

$$x_{n+1} = x_n - \frac{x_n^3 - a}{3x_n^2}$$

Step 5. [1] Split the fraction into two pieces and simplify:

$$x_{n+1} = x_n - \frac{x_n^3}{3x_n^2} + \frac{a}{3x_n^2} = x_n - \frac{x_n}{3} + \frac{a}{3x_n^2} = \frac{2x_n}{3} + \frac{a}{3x_n^2}$$

(c) Secant for $\sqrt{a}$ (2025 Q11)

Step 6. [1] Choose $f(x) = x^2 - a$.

Step 7. [2] Apply the secant method. In the denominator the a's cancel, leaving a difference of squares:

$$x_{n+1} = x_n - (x_n^2 - a)\,\frac{x_n - x_{n-1}}{(x_n^2 - a) - (x_{n-1}^2 - a)} = x_n - (x_n^2 - a)\,\frac{x_n - x_{n-1}}{x_n^2 - x_{n-1}^2}$$

Step 8. [1] Factor $x_n^2 - x_{n-1}^2 = (x_n - x_{n-1})(x_n + x_{n-1})$ and cancel $(x_n - x_{n-1})$:

$$x_{n+1} = x_n - \frac{x_n^2 - a}{x_n + x_{n-1}} = x_n + \frac{a - x_n^2}{x_n + x_{n-1}}$$

(d) Secant for $\sqrt[3]{a}$ (Practice Exam 2)

Step 9. Same as (c) with $f(x) = x^3 - a$, using the difference of cubes $x_n^3 - x_{n-1}^3 = (x_n - x_{n-1})(x_n^2 + x_n x_{n-1} + x_{n-1}^2)$:

$$x_{n+1} = x_n - \frac{x_n^3 - a}{x_n^2 + x_n x_{n-1} + x_{n-1}^2}$$

(e) Newton for $\sqrt[4]{a}$ (Practice Exam 1)

Step 10. $f(x) = x^4 - a$, $f'(x) = 4x^3$: $\quad x_{n+1} = x_n - \dfrac{x_n^4 - a}{4x_n^3} = \dfrac{3x_n}{4} + \dfrac{a}{4x_n^3}$.

Result. the recipe is always: $f = x^k - a \to$ apply the method $\to$ split the fraction (Newton) or factor and cancel (secant).

D9.4 Why Newton converges quadratically

Priority: ★ explains "the correct digits double"

Goal: show $x_{n+1} - p \approx C\,(x_n - p)^2$, where p is the root.

Idea. Taylor-expand f about x_n and evaluate it at the root, where $f(p) = 0$.

Step 1. Taylor degree 1 with remainder, centred on x_n, evaluated at the root p (where $f(p) = 0$):

$$0 = f(p) = f(x_n) + f'(x_n)(p - x_n) + \frac{f''(\xi)}{2}(p - x_n)^2$$

Step 2. Divide by $f'(x_n)$ and rearrange:

$$\underbrace{x_n - \frac{f(x_n)}{f'(x_n)}}_{=x_{n+1}} - p = \frac{f''(\xi)}{2f'(x_n)}\,(x_n - p)^2$$

Step 3. So the new error is (a constant) $\times$ (the old error)^2: if the error is 10^{-3}, the next is roughly 10^{-6}.

Result. $|x_{n+1} - p| \approx \left|\dfrac{f''(p)}{2f'(p)}\right| |x_n - p|^2$: quadratic convergence. It breaks down if $f'(p) = 0$ (a repeated root), where Newton is only linear.

Weeks 10–11: Numerical differentiation and integration

Notation. $f_0 = f(x_0)$, $f_0' = f'(x_0)$, $f_0'' = f''(x_0)$, and so on. All the differentiation formulas come from Toolkit T1 (Taylor with step h).

D10.1 Forward and backward differences, with error terms

Priority: ★★

Goal: derive $f'(x_0) = \dfrac{f(x_0+h)-f(x_0)}{h} - \dfrac{h}{2}f''(c)$, and the backward version.

Idea. a degree-1 Taylor expansion contains f'. Solve for it.

Step 1. Taylor degree 1 with remainder (T1), c between x_0 and x_0+h:

$$f(x_0+h) = f_0 + h\,f_0' + \frac{h^2}{2}f''(c)$$

Step 2. Move f_0 to the left:

$$f(x_0+h) - f_0 = h\,f_0' + \frac{h^2}{2}f''(c)$$

Step 3. Divide by h:

$$\frac{f(x_0+h)-f_0}{h} = f_0' + \frac{h}{2}f''(c)$$

Step 4. Rearrange for f_0':

$$f'(x_0) = \frac{f(x_0+h)-f(x_0)}{h} - \frac{h}{2}f''(c) \qquad \text{(forward, error } O(h)\text{)}$$

Step 5. Backward: replace h by $-h$ everywhere, then tidy the signs (multiply the top and bottom of the fraction by -1):

$$f'(x_0) = \frac{f(x_0-h)-f(x_0)}{-h} - \frac{(-h)}{2}f''(c) = \frac{f(x_0)-f(x_0-h)}{h} + \frac{h}{2}f''(c)$$

Result. forward: $\dfrac{f(x_0+h)-f(x_0)}{h}$; backward: $\dfrac{f(x_0)-f(x_0-h)}{h}$. Both are first order, $O(h)$, with errors of opposite sign.

D10.2 Central difference for f', with error term

Priority: ★★

Goal: derive $f'(x_0) = \dfrac{f(x_0+h)-f(x_0-h)}{2h} - \dfrac{h^2}{6}f'''(c)$.

Idea. expand $f(x_0+h)$ and $f(x_0-h)$, then **subtract**: the even-power terms (f_0, f_0'') cancel, leaving f_0'.

Step 1. Degree-2 expansions with remainders (T1), $c_1 \in (x_0, x_0+h)$, $c_2 \in (x_0-h, x_0)$:

$$f(x_0+h) = f_0 + hf_0' + \frac{h^2}{2}f_0'' + \frac{h^3}{6}f'''(c_1) \qquad (1)$$

$$f(x_0 - h) = f_0 - hf_0' + \frac{h^2}{2}f_0'' - \frac{h^3}{6}f'''(c_2) \qquad (2)$$

Step 2. Compute (1) − (2). The f_0 and $\frac{h^2}{2}f_0''$ terms cancel:

$$f(x_0 + h) - f(x_0 - h) = 2hf_0' + \frac{h^3}{6}\left[f'''(c_1) + f'''(c_2)\right]$$

Step 3. Divide by $2h$ and rearrange for f_0':

$$f'(x_0) = \frac{f(x_0 + h) - f(x_0 - h)}{2h} - \frac{h^2}{6} \cdot \frac{f'''(c_1) + f'''(c_2)}{2}$$

Step 4. Merge the two unknown points with the IVT trick (T3): the average equals $f'''(c)$ for some c between c_1 and c_2.

Result. $f'(x_0) = \dfrac{f(x_0 + h) - f(x_0 - h)}{2h} - \dfrac{h^2}{6}f'''(c), \quad c \in (x_0 - h, x_0 + h)$. Second order, $O(h^2)$.

D10.3 Central difference for f'', with error term

Priority: ★★★ 2023 Q11a (4 marks)

Goal: derive $f''(x_0) = \dfrac{f(x_0 - h) - 2f(x_0) + f(x_0 + h)}{h^2} - \dfrac{h^2}{12}f^{(4)}(c), \quad c \in (x_0 - h, x_0 + h)$.

Idea. expand $f(x_0 \pm h)$ up to h^3, then **add**: the odd-power terms (f_0', f_0''') cancel, leaving f_0''.

Step 1. [1] Degree-3 expansions with h^4 remainders, $c_1 \in (x_0, x_0 + h)$, $c_2 \in (x_0 - h, x_0)$:

$$f(x_0 + h) = f_0 + hf_0' + \frac{h^2}{2}f_0'' + \frac{h^3}{6}f_0''' + \frac{h^4}{24}f^{(4)}(c_1) \qquad (1)$$

$$f(x_0 - h) = f_0 - hf_0' + \frac{h^2}{2}f_0'' - \frac{h^3}{6}f_0''' + \frac{h^4}{24}f^{(4)}(c_2) \qquad (2)$$

Why: we go up to h^3 because, after adding and dividing by h^2, we need to see the size of the error.

Step 2. [1] Add (1) + (2). The $\pm hf_0'$ and $\pm\frac{h^3}{6}f_0'''$ terms cancel:

$$f(x_0 + h) + f(x_0 - h) = 2f_0 + h^2 f_0'' + \frac{h^4}{24}\left[f^{(4)}(c_1) + f^{(4)}(c_2)\right]$$

Step 3. [1] Rearrange for f_0'': subtract $2f_0$, then divide by h^2:

$$f''(x_0) = \frac{f(x_0 - h) - 2f(x_0) + f(x_0 + h)}{h^2} - \frac{h^2}{24}\left[f^{(4)}(c_1) + f^{(4)}(c_2)\right]$$
$$= \frac{f(x_0 - h) - 2f(x_0) + f(x_0 + h)}{h^2} - \frac{h^2}{12} \cdot \frac{f^{(4)}(c_1) + f^{(4)}(c_2)}{2}$$

Step 4. [1] IVT (T3): provided $f^{(4)}$ is continuous on $[x_0 - h, x_0 + h]$, there is a c between c_1 and c_2 with $f^{(4)}(c) = \dfrac{f^{(4)}(c_1) + f^{(4)}(c_2)}{2}$.

Result. $f''(x_0) = \dfrac{f(x_0 - h) - 2f(x_0) + f(x_0 + h)}{h^2} - \dfrac{h^2}{12}f^{(4)}(c)$. Second order, $O(h^2)$.

Common mistake: forgetting to *state* the IVT step. It is worth a mark on its own.

D10.4 Second-order forward difference

Priority: ★★ lecture problem

Goal: derive $f'(x_0) \approx \dfrac{-3f(x_0)+4f(x_0+h)-f(x_0+2h)}{2h}$ with error $O(h^2)$.

Idea. use two points to the right. Combine their expansions so the f'' terms cancel.

Step 1. Expand both up to h^3 (for x_0+2h, put $2h$ in place of h: $\frac{(2h)^2}{2}=2h^2$ and $\frac{(2h)^3}{6}=\frac{4}{3}h^3$):

$$f(x_0+h) = f_0 + hf_0' + \frac{h^2}{2}f_0'' + \frac{h^3}{6}f_0''' + O(h^4) \qquad (1)$$

$$f(x_0+2h) = f_0 + 2hf_0' + 2h^2 f_0'' + \frac{4h^3}{3}f_0''' + O(h^4) \qquad (2)$$

Step 2. Choose a combination that kills f_0'': (1) has $\frac{h^2}{2}f_0''$ and (2) has $2h^2 f_0''$, so take $4\times(1)-(2)$:

$$4f(x_0+h)-f(x_0+2h) = 3f_0 + 2hf_0' + \left(\frac{4}{6}-\frac{4}{3}\right)h^3 f_0''' + O(h^4) = 3f_0 + 2hf_0' - \frac{2}{3}h^3 f_0''' + O(h^4)$$

Step 3. Rearrange for f_0': move $3f_0$ across, then divide by $2h$:

$$f'(x_0) = \frac{-3f(x_0)+4f(x_0+h)-f(x_0+2h)}{2h} + \frac{h^2}{3}f_0''' + O(h^3)$$

Result. second-order accuracy using only points to the right (useful at the left end of a table).

D10.5 How the choice of h affects accuracy

Priority: ★★★ 2023 Q11b (2 marks) + ★ the calculation

Goal: explain why h can be too large *or* too small.

Step 1. [1 + 1] **Exam answer:** "Too large a value of h gives inaccuracy from **truncation error** (the Taylor terms we dropped, $O(h^p)$). Too small a value gives inaccuracy from **roundoff error**: the numerator subtracts two nearly-equal numbers (catastrophic cancellation), and dividing by a tiny h magnifies it."

Step 2. **(Understanding) Model the forward difference's total error.** Each stored value of f has relative rounding error up to ε, so the numerator can be off by about $2\varepsilon|f|$. After dividing by h:

$$E(h) \approx \underbrace{\frac{M}{2}h}_{\text{truncation}} + \underbrace{\frac{2\varepsilon|f|}{h}}_{\text{roundoff}} \qquad (M \approx |f''|)$$

Step 3. Minimise: set $E'(h) = \frac{M}{2} - \frac{2\varepsilon|f|}{h^2} = 0$, giving $h = 2\sqrt{\varepsilon|f|/M}$. With $|f| \approx M$, this is about $\sqrt{\varepsilon} \approx 10^{-8}$ in double precision.

Result. for the forward difference the best h is about $\sqrt{\varepsilon} \approx 10^{-8}$ (the bottom of the V-shaped error plot). For the central difference it is about $\varepsilon^{1/3} \approx 10^{-5}$.

D11.1 Trapezoidal rule for one strip

Priority: ★★★ 2023 Q13a (3 marks)

Goal: show $\displaystyle\int_{x_0}^{x_1} P_1(x)\,dx = \frac{h}{2}\big[f(x_0)+f(x_1)\big]$, where P_1 interpolates f at x_0 and x_1.

Idea. write the straight line in Newton forward difference form, switch to s (T4), then integrate a simple polynomial in s.

Step 1. [1] The degree-1 interpolant in forward difference form:

$$P_1(x) = y_0 + s\,\Delta y_0, \qquad s = \frac{x-x_0}{h}, \qquad \Delta y_0 = y_1 - y_0$$

Step 2. Change variable (T4): $x = x_0 + sh$ gives $dx = h\,ds$. At $x = x_0$, $s = 0$; at $x = x_1$, $s = 1$.

Step 3. [1] Integrate in s:

$$\int_{x_0}^{x_1} P_1\,dx = \int_0^1 \big(y_0 + s\,\Delta y_0\big)\,h\,ds = h\Big[y_0 s + \tfrac{1}{2}s^2\Delta y_0\Big]_0^1 = h\Big(y_0 + \tfrac{1}{2}\Delta y_0\Big)$$

Step 4. [1] Substitute $\Delta y_0 = y_1 - y_0$ and simplify:

$$h\Big(y_0 + \tfrac{1}{2}(y_1 - y_0)\Big) = h\Big(\tfrac{1}{2}y_0 + \tfrac{1}{2}y_1\Big) = \frac{h}{2}\big[f(x_0)+f(x_1)\big]$$

Result. $\displaystyle\int_{x_0}^{x_1} f\,dx \approx \frac{h}{2}\big[f(x_0)+f(x_1)\big]$, which is the area of a trapezoid.

D11.2 Composite trapezoidal rule

Priority: ★★★ 2023 Q13b (2 marks)

Goal: show $\displaystyle\int_a^b f\,dx \approx \frac{h}{2}\Big[f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\Big]$, with $h = \frac{b-a}{n}$.

Idea. split $[a,b]$ into n strips, apply D11.1 to each, and add.

Step 1. [1] Split the integral into strips:

$$\int_a^b f\,dx = \int_{x_0}^{x_1} f\,dx + \int_{x_1}^{x_2} f\,dx + \cdots + \int_{x_{n-1}}^{x_n} f\,dx$$

Step 2. [0.5] Apply D11.1 to every strip:

$$\approx \frac{h}{2}\big[f_0 + f_1\big] + \frac{h}{2}\big[f_1 + f_2\big] + \cdots + \frac{h}{2}\big[f_{n-1} + f_n\big]$$

Step 3. [0.5] Collect terms. Each interior point $x_1, \ldots, x_{n-1}$ is the right end of one strip and the left end of the next, so it appears **twice**. The endpoints appear once:

$$= \frac{h}{2}\Big[f(x_0) + 2\big(f(x_1) + \cdots + f(x_{n-1})\big) + f(x_n)\Big]$$

Result. composite trapezoidal rule, with weights $1, 2, 2, \ldots, 2, 1$. Error $-\frac{(b-a)h^2}{12}f''(c)$, so second order.

D11.3 Simpson's rule for one pair of strips

Priority: ★★★ 2024 Q13 (5 marks)

Goal: show $\displaystyle\int_{x_0}^{x_2} P_2(x)\,dx = \frac{h}{3}\big[f(x_0)+4f(x_1)+f(x_2)\big]$.

Idea. same as D11.1, but with the degree-2 interpolant and s running from 0 to 2.

Step 1. [1] The degree-2 interpolant in forward difference form. Expand $\frac{s(s-1)}{2} = \frac{1}{2}s^2 - \frac{1}{2}s$ ready for integrating:

$$P_2(x) = y_0 + s\,\Delta y_0 + \left(\tfrac{1}{2}s^2 - \tfrac{1}{2}s\right)\Delta^2 y_0$$

Step 2. Change variable: $dx = h\,ds$. At $x = x_0$, $s = 0$; at $x = x_2$, $s = 2$.

Step 3. [1] Integrate term by term:

$$\int_{x_0}^{x_2} P_2\,dx = h\Big[y_0 s + \tfrac{1}{2}s^2\Delta y_0 + \left(\tfrac{1}{6}s^3 - \tfrac{1}{4}s^2\right)\Delta^2 y_0\Big]_0^2$$

Step 4. Evaluate at $s = 2$ (everything is 0 at $s = 0$). Note that $\frac{1}{6}(8) - \frac{1}{4}(4) = \frac{4}{3} - 1 = \frac{1}{3}$:

$$= h\Big[2y_0 + 2\Delta y_0 + \tfrac{1}{3}\Delta^2 y_0\Big]$$

Step 5. [1] Substitute $\Delta y_0 = y_1 - y_0$ and $\Delta^2 y_0 = y_2 - 2y_1 + y_0$:

$$= h\Big[2y_0 + 2y_1 - 2y_0 + \tfrac{1}{3}y_2 - \tfrac{2}{3}y_1 + \tfrac{1}{3}y_0\Big]$$

Step 6. [1] Collect each y: y_0: $2 - 2 + \frac{1}{3} = \frac{1}{3}$; y_1: $2 - \frac{2}{3} = \frac{4}{3}$; y_2: $\frac{1}{3}$:

$$= h\Big[\tfrac{1}{3}y_0 + \tfrac{4}{3}y_1 + \tfrac{1}{3}y_2\Big]$$

Step 7. [1] Factor out $\frac{1}{3}$:

$$= \frac{h}{3}\big[f(x_0)+4f(x_1)+f(x_2)\big]$$

Result. $\displaystyle\int_{x_0}^{x_2} f\,dx \approx \frac{h}{3}\big[f(x_0)+4f(x_1)+f(x_2)\big]$, the area under the parabola through the three points.

D11.4 Composite Simpson's rule

Priority: ★★

Goal: show $\displaystyle\int_a^b f\,dx \approx \frac{h}{3}\Big[f_0 + 4\sum_{\text{odd } i} f_i + 2\sum_{\text{even interior } i} f_i + f_n\Big]$, with n even.

Idea. apply D11.3 to consecutive *pairs* of strips, then add. That's why n must be even.

Step 1. Split into pairs: $[x_0, x_2], [x_2, x_4], \ldots, [x_{n-2}, x_n]$. There are $\frac{n}{2}$ pairs, so n must be even.

Step 2. Apply D11.3 to each pair:

$$\frac{h}{3}\big[f_0 + 4f_1 + f_2\big] + \frac{h}{3}\big[f_2 + 4f_3 + f_4\big] + \cdots + \frac{h}{3}\big[f_{n-2} + 4f_{n-1} + f_n\big]$$

Step 3. Collect terms. The **odd** points are pair middles, so they get weight 4. Each **even interior** point is shared by two pairs $(1+1)$, so it gets weight 2. The ends get weight 1.

Result. weights $1, 4, 2, 4, \ldots, 2, 4, 1$, multiplied by $\frac{h}{3}$. Error $-\frac{(b-a)h^4}{180} f^{(4)}(c)$, so fourth order.

D11.5 Why Trapezoid is exact for lines and Simpson is exact for cubics

Priority: ★

Goal: see where the "exact for degree ≤ 1" and "exact for degree ≤ 3" facts come from.

Step 1. From the error terms. The trapezoid's error contains f'', and $f'' = 0$ for any straight line, so the error is 0. Simpson's error contains $f^{(4)}$, and $f^{(4)} = 0$ for any cubic.

Step 2. A direct check that Simpson is exact for x^3. On $[-h, h]$ (nodes $-h, 0, h$):

$$\int_{-h}^{h} x^3\,dx = 0 \quad \text{(odd function)}, \qquad \frac{h}{3}\left[(-h)^3 + 4\cdot 0 + h^3\right] = 0 \ 3$$

Step 3. And for x^2:

$$\int_{-h}^{h} x^2\,dx = \frac{2h^3}{3}, \qquad \frac{h}{3}\left[h^2 + 4\cdot 0 + h^2\right] = \frac{2h^3}{3} \ 3$$

Result. Simpson fits a parabola but is exact one degree higher, for cubics: the symmetric weights cancel the x^3 error.

Weeks 12–13: Linear systems

D12.1 Forward substitution

Priority: ★★

Goal: solve a lower triangular system $A\mathbf{x} = \mathbf{b}$ (all zeros above the diagonal).

Idea. row 1 contains only x_1. Once x_1 is known, row 2 contains only one new unknown, and so on downwards.

Step 1. Row i of a lower triangular system only involves $x_1, \ldots, x_i$ (the entries a_{ij} with $j > i$ are 0):

$$a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{i,i-1}x_{i-1} + a_{ii}x_i = b_i$$

Step 2. Move the already-known terms to the right-hand side, then divide by the diagonal entry:

$$x_i = \frac{b_i - \sum_{j=1}^{i-1} a_{ij}x_j}{a_{ii}}, \qquad i = 1, 2, \ldots, n \quad \text{(top to bottom)}$$

Result. forward substitution. It needs every $a_{ii} \neq 0$.

D12.2 Backward substitution

Priority: ★★

Goal: solve an upper triangular system (all zeros below the diagonal).

Idea. the last row contains only x_n. Work upwards.

Step 1. Row i only involves $x_i, \ldots, x_n$: $\quad a_{ii}x_i + a_{i,i+1}x_{i+1} + \cdots + a_{in}x_n = b_i$.

Step 2. Move the known terms (those to the right of the diagonal) across, then divide:

$$x_i = \frac{b_i - \sum_{j=i+1}^{n} a_{ij}x_j}{a_{ii}}, \qquad i = n, n-1, \ldots, 1 \quad \text{(bottom to top)}$$

Result. backward substitution. It needs every $a_{ii} \neq 0$.

D12.3 How LU factorisation solves $A\mathbf{x} = \mathbf{b}$

Priority: ★★★ 2023 Q15b (3 marks)

Goal: explain, in words and symbols, how $A = LU$ is used to solve $A\mathbf{x} = \mathbf{b}$.

Idea. one hard system becomes two easy triangular ones.

Step 1. [1] Replace A by its factorisation, and regroup the brackets (matrix multiplication is associative):

$$A\mathbf{x} = \mathbf{b} \;\Rightarrow\; (LU)\mathbf{x} = \mathbf{b} \;\Rightarrow\; L\,(U\mathbf{x}) = \mathbf{b}$$

Step 2. Name the bracket $\mathbf{z} = U\mathbf{x}$. Then the system reads $L\mathbf{z} = \mathbf{b}$.

Step 3. [1] Solve $L\mathbf{z} = \mathbf{b}$ for $\mathbf{z}$ by **forward substitution** (L is lower triangular).

Step 4. [1] Solve $U\mathbf{x} = \mathbf{z}$ for $\mathbf{x}$ by **backward substitution** (U is upper triangular).

Result. $A\mathbf{x} = \mathbf{b} \to L\mathbf{z} = \mathbf{b}$ (forward) $\to U\mathbf{x} = \mathbf{z}$ (backward). For a new $\mathbf{b}$, reuse the same L and U.

D12.4 Doolittle LU formulas (3×3)

Priority: ★★★ Q15 every year (2–3 marks for the missing entries)

Goal: find formulas for the entries of L (with 1s on its diagonal) and U such that $A = LU$.

Idea. multiply L and U out symbolically, set each entry equal to the matching entry of A, and solve in an order where only one entry is unknown at a time.

Step 1. Write out the factors:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}, \qquad U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Step 2. Multiply them: entry (i, j) is (row i of L) $\cdot$ (column j of U):

$$LU = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ \ell_{21}u_{11} & \ell_{21}u_{12} + u_{22} & \ell_{21}u_{13} + u_{23} \\ \ell_{31}u_{11} & \ell_{31}u_{12} + \ell_{32}u_{22} & \ell_{31}u_{13} + \ell_{32}u_{23} + u_{33} \end{bmatrix}$$

Step 3. Set $LU = A$ entry by entry and solve in this order: **row 1 of U**, then **column 1 of L**, then **row 2 of U**, then **column 2 of L**, then u_{33}:

row 1 of U :	$u_{11} = a_{11}, \quad u_{12} = a_{12}, \quad u_{13} = a_{13}$
column 1 of L :	$\ell_{21} = \dfrac{a_{21}}{u_{11}}, \quad \ell_{31} = \dfrac{a_{31}}{u_{11}}$
row 2 of U :	$u_{22} = a_{22} - \ell_{21}u_{12}, \quad u_{23} = a_{23} - \ell_{21}u_{13}$
column 2 of L :	$\ell_{32} = \dfrac{a_{32} - \ell_{31}u_{12}}{u_{22}}$
last entry :	$u_{33} = a_{33} - \ell_{31}u_{13} - \ell_{32}u_{23}$

Why: in this order, each equation contains exactly one entry you haven't found yet.

Worked example (2024 Q15a). *What the question gives you:* A, and a partly filled-in L and U. The boxed entries are the blanks:

$$\underbrace{\begin{bmatrix} 2 & 1 & -4 \\ -4 & -5 & 10 \\ 6 & 6 & -10 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & \boxed{\ell_{32}} & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 2 & 1 & -4 \\ 0 & \boxed{u_{22}} & 2 \\ 0 & 0 & \boxed{u_{33}} \end{bmatrix}}_{U}$$

Your job: find the three boxed numbers. For each one, take the row of L and column of U that contain it, set their dot product equal to the matching entry of A, and solve. (The formulas from Step 3 give the same thing.)

Step 4. **Find** u_{22} using row 2 of L and column 2 of U, which must give $a_{22} = -5$:

$$(-2,\ 1,\ 0)\cdot(1,\ u_{22},\ 0) = -2 + u_{22} = -5 \ \Rightarrow\ u_{22} = -3 \qquad \Big(\text{formula: } a_{22} - \ell_{21}u_{12} = -5 - (-2)(1)\Big)$$

Step 5. **Find** ℓ_{32} using row 3 of L and column 2 of U (with $u_{22} = -3$ now filled in), which must give $a_{32} = 6$:

$$(3,\ \ell_{32},\ 1)\cdot(1,\ -3,\ 0) = 3-3\ell_{32} = 6 \ \Rightarrow\ \ell_{32} = -1 \qquad \left(\text{formula: } \tfrac{a_{32}-\ell_{31}u_{12}}{u_{22}} = \tfrac{6-3}{-3}\right)$$

Step 6. **Find** u_{33} using row 3 of L (with $\ell_{32} = -1$) and column 3 of U, which must give $a_{33} = -10$:

$$(3,\ -1,\ 1)\cdot(-4,\ 2,\ u_{33}) = -12-2+u_{33} = -10 \ \Rightarrow\ u_{33} = 4 \qquad \left(\text{formula: } a_{33}-\ell_{31}u_{13}-\ell_{32}u_{23}\right)$$

Check: row 3 of L times U: $\big(3\cdot 2,\ 3\cdot 1+(-1)(-3),\ -10\big) = (6,6,-10)$, which is row 3 of A ✓
Why: order matters: u_{22} must come before ℓ_{32}, because the ℓ_{32} equation uses u_{22}.

Result. every Doolittle formula comes from one rule: $a_{ij} = (\text{row } i \text{ of } L)\cdot(\text{column } j \text{ of } U)$.

Common mistake: a zero pivot ($u_{11} = 0$ or $u_{22} = 0$) makes a division impossible. Pivoting (swapping rows) fixes it.

D13.1 Showing a matrix is positive definite

Priority: ★★★ 2025 Q16a (3 marks)

Goal: show $A = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$ is positive definite, i.e. $\mathbf{x}^T A\mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0}$.

Idea. multiply $\mathbf{x}^T A\mathbf{x}$ out into a quadratic, then complete the square to show it's a sum of squares.

Step 1. [1] State what has to be shown: $\mathbf{x}^T A\mathbf{x} > 0$ for all nonzero $\mathbf{x} = (x_1, x_2)^T$.

Step 2. [1] Multiply out, starting with $A\mathbf{x}$:

$$A\mathbf{x} = \begin{bmatrix} x_1 - x_2 \\ -x_1 + 2x_2 \end{bmatrix}, \qquad \mathbf{x}^T A\mathbf{x} = x_1(x_1 - x_2) + x_2(-x_1 + 2x_2) = x_1^2 - 2x_1x_2 + 2x_2^2$$

Step 3. [0.5] Complete the square. Split $2x_2^2 = x_2^2 + x_2^2$ so that a perfect square appears:

$$x_1^2 - 2x_1x_2 + x_2^2 + x_2^2 = (x_1 - x_2)^2 + x_2^2$$

Step 4. [0.5] Conclude. A sum of squares is ≥ 0. It equals 0 only if $x_2 = 0$ *and* $x_1 = x_2$, which means $\mathbf{x} = \mathbf{0}$. So $\mathbf{x}^T A\mathbf{x} > 0$ for every nonzero $\mathbf{x}$, and A is positive definite.

General 2×2 version ★. For $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ with $a > 0$:

$$\mathbf{x}^T A\mathbf{x} = ax_1^2 + 2bx_1x_2 + cx_2^2 = a\Big(x_1 + \frac{b}{a}x_2\Big)^2 + \Big(c - \frac{b^2}{a}\Big)x_2^2$$

so A is positive definite exactly when $a > 0$ and $ac - b^2 > 0$. **Check:** for $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$: $a = 1 > 0$ and $ac - b^2 = 2 - 1 = 1 > 0$ ✓

Result. to prove positive definiteness: multiply out $\mathbf{x}^T A\mathbf{x}$, write it as a sum of squares, and argue it is 0 only for $\mathbf{x} = \mathbf{0}$.

D13.2 Cholesky formulas (3×3)

Priority: ★★★ Q16 every year (2–3 marks for the missing entries)

Goal: find the entries of the lower triangular L with $A = LL^T$, for a symmetric positive definite A.

Idea. exactly like D12.4: multiply out, equate entries with A, and solve column by column.

Step 1. Write the factors (L^T is L reflected across its diagonal):

$$L = \begin{bmatrix} \ell_{11} & 0 & 0 \\ \ell_{21} & \ell_{22} & 0 \\ \ell_{31} & \ell_{32} & \ell_{33} \end{bmatrix}, \qquad L^T = \begin{bmatrix} \ell_{11} & \ell_{21} & \ell_{31} \\ 0 & \ell_{22} & \ell_{32} \\ 0 & 0 & \ell_{33} \end{bmatrix}$$

Step 2. Multiply. Only the lower triangle is needed, because A is symmetric:

$$LL^T = \begin{bmatrix} \ell_{11}^2 & \cdot & \cdot \\ \ell_{21}\ell_{11} & \ell_{21}^2 + \ell_{22}^2 & \cdot \\ \ell_{31}\ell_{11} & \ell_{31}\ell_{21} + \ell_{32}\ell_{22} & \ell_{31}^2 + \ell_{32}^2 + \ell_{33}^2 \end{bmatrix}$$

Step 3. Set this equal to A and solve **column by column**:

$$\begin{aligned} &\text{column 1}: && \ell_{11} = \sqrt{a_{11}}, \quad \ell_{21} = \frac{a_{21}}{\ell_{11}}, \quad \ell_{31} = \frac{a_{31}}{\ell_{11}} \\ &\text{column 2}: && \ell_{22} = \sqrt{a_{22} - \ell_{21}^2}, \quad \ell_{32} = \frac{a_{32} - \ell_{31}\ell_{21}}{\ell_{22}} \\ &\text{column 3}: && \ell_{33} = \sqrt{a_{33} - \ell_{31}^2 - \ell_{32}^2} \end{aligned}$$

Why: a diagonal entry comes from a sum of squares, so it needs a square root (take the positive one). An entry below the diagonal comes from a dot product, so it needs a division.

Worked example (2025 Q16b). *What the question gives you:* A, and a partly filled-in L (with L^T beside it). The boxed entries are the blanks:

$$\underbrace{\begin{bmatrix} 36 & -12 & -6 \\ -12 & 13 & 11 \\ -6 & 11 & 35 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} \boxed{\ell_{11}} & 0 & 0 \\ -2 & 3 & 0 \\ -1 & \boxed{\ell_{32}} & \boxed{\ell_{33}} \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} \ell_{11} & -2 & -1 \\ 0 & 3 & \ell_{32} \\ 0 & 0 & \ell_{33} \end{bmatrix}}_{L^T}$$

Your job: find the three boxed numbers. Same idea as LU: (row of L) $\cdot$ (column of L^T) = entry of A. Column j of L^T is just row j of L, so entry a_{ij} is (row i of L) $\cdot$ (row j of L).

Step 4. Find ℓ_{11} using row 1 $\cdot$ row 1, which must give $a_{11} = 36$:

$$(\ell_{11}, 0, 0) \cdot (\ell_{11}, 0, 0) = \ell_{11}^2 = 36 \;\Rightarrow\; \ell_{11} = 6 \qquad \text{(take the positive root)}$$

Step 5. Find ℓ_{32} using row 3 $\cdot$ row 2, which must give $a_{32} = 11$:

$$(-1,\ \ell_{32},\ \ell_{33}) \cdot (-2,\ 3,\ 0) = 2 + 3\ell_{32} = 11 \;\Rightarrow\; \ell_{32} = 3 \qquad \left(\text{formula: } \tfrac{a_{32} - \ell_{31}\ell_{21}}{\ell_{22}} = \tfrac{11-2}{3}\right)$$

Step 6. Find ℓ_{33} using row 3 $\cdot$ row 3, which must give $a_{33} = 35$:

$$(-1,\ 3,\ \ell_{33}) \cdot (-1,\ 3,\ \ell_{33}) = 1 + 9 + \ell_{33}^2 = 35 \;\Rightarrow\; \ell_{33}^2 = 25 \;\Rightarrow\; \ell_{33} = 5$$

Check: row 2 $\cdot$ row 2 $= (-2)^2 + 3^2 = 13 = a_{22}$ ✓, row 2 $\cdot$ row 1 $= (-2)(6) = -12 = a_{21}$ ✓

Result. diagonal: $\sqrt{\text{entry} - \text{sum of squares so far}}$; below the diagonal: (entry − dot product so far) ÷ diagonal.

Common mistake: a negative (or zero) number under a square root means A is **not** positive definite. The factorisation fails, so switch to LU.

D13.3 How Cholesky solves $A\mathbf{x} = \mathbf{b}$

Priority: ★★

Goal: solve $A\mathbf{x} = \mathbf{b}$ using $A = LL^T$.

Step 1. Replace A and regroup: $(LL^T)\mathbf{x} = \mathbf{b} \;\Rightarrow\; L\,(L^T\mathbf{x}) = \mathbf{b}$.

Step 2. Let $\mathbf{z} = L^T\mathbf{x}$. Solve $L\mathbf{z} = \mathbf{b}$ by **forward** substitution.

Step 3. Solve $L^T\mathbf{x} = \mathbf{z}$ by **backward** substitution (L^T is upper triangular).

Result. the same two-step idea as LU, with $U = L^T$. It is about half the work, needs no pivoting, and succeeds exactly when A is symmetric positive definite.

Final tip. Many of these derivations share a skeleton:

- ODE methods: Taylor about t_i → evaluate at t_{i+1} → $y' = f$ → drop $O(h^{\cdot})$.
- Finite differences: expand $f(x_0 \pm h)$ → add or subtract → rearrange → IVT.
- Quadrature: interpolant in s → $dx = h\,ds$ → integrate → substitute the Δ's.
- Matrix factorisations: multiply out → equate entries → solve one unknown at a time.

Learn the skeletons, and the details become much easier to remember.